

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – DEBUGGING & OPTIMISATION TASK

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Debugging and Optimisation task
Weightage:	45%	Total Marks:	100
CO2 Statement:	Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems.		
Student Name:	ASWIN M	Reg. No.:	192412437
Section / Batch:	B.E (ECE) 3 rd Year	Date:	04/08/2026

Instructions to Students

- Identify arithmetic errors and performance bottlenecks in the given problem.
- Analyse the execution steps to locate the source of errors.
- Debug the algorithm by correcting logical and computational faults.
- Optimize the solution using appropriate arithmetic techniques or algorithms.
- Evaluate the optimized solution for correctness, accuracy, and efficiency.

QUESTIONS

1. A digital signal processing system produces incorrect results during signed 8-bit integer addition using 2's complement representation when large positive numbers are added. Debug the binary addition process to determine whether overflow has occurred. Recommend an effective overflow detection and handling mechanism to ensure reliable computation.
2. A high-speed embedded controller experiences increased execution time because its arithmetic unit is built with a ripple carry adder. Debug the carry propagation process to identify the cause of the delay. Propose the use of a carry look-ahead adder and explain how it improves the overall arithmetic performance.
3. A processor's signed multiplication module generates incorrect outputs when multiplying negative binary numbers using Booth's algorithm. Debug the algorithm by examining bit-pair evaluation, arithmetic right shifts, and sign extension at each iteration. Identify the fault and recommend modifications to produce accurate multiplication results.
4. A microcontroller performing binary division exhibits poor performance because its restoring division implementation repeatedly restores the partial remainder after unsuccessful subtraction. Debug the sequence of operations to identify redundant steps. Suggest replacing it with the non-restoring division algorithm and explain how the optimization reduces execution time.
5. A numerical computing application produces inaccurate floating-point results when combining values with widely different magnitudes using IEEE 754 single-precision representation. Debug the computation by examining exponent alignment, normalization, and rounding operations. Recommend suitable optimization techniques, such as higher precision or improved numerical methods, to enhance computational accuracy.

Code of Conduct

I ASWIN M - 192412437(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Identification of Errors / Bugs	20	Accurately identifies all errors and issues with complete understanding of the problem	Identifies most errors with minor omissions	Identifies limited errors; misses key issues	Unable to identify errors or misunderstands the problem
Debugging Approach & Analysis	20	Uses effective debugging techniques with clear analysis and root cause identification	Applies suitable debugging methods with minor gaps	Uses basic debugging methods with limited analysis	No systematic debugging approach followed
Correctness of Debugged Solution	25	Provides completely corrected solution with accurate functionality and expected output	Provides working solution with minor errors	Partially correct solution with limited functionality	Incorrect solution or fails to resolve errors
Optimization Techniques Applied	20	Applies efficient optimization methods to improve performance, memory usage and quality	Performs optimization with noticeable improvements	Limited optimization with minimal improvements	No optimization applied or inefficient implementation
Testing and Documentation	15	Performs complete testing with proper validation and clear documentation	Provides adequate testing and documentation with minor issues	Limited testing and incomplete documentation	No proper testing or documentation provided

Marks Evaluation:

Q. No	Identification of Errors / Bugs (20)	Debugging Approach & Analysis (20)	Correctness of Debugged Solution (25)	Optimization Techniques Applied (20)	Testing and Documentation (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

CO2 Assesment Tool - 2

Name: Aswin M
Reg No: 192412437
Subject Code:
CSA1220

Question 1

Given

Number 1 = +100

Number 2 = +50

Convert to 8-bit Binary

+100 = 01100100

+50 = 00110010

Binary Addition

01100100

+ 00110010

10010110

Decimal Verification

Expected Result:

100 + 50 = 150

Obtained Binary Result:

10010110

Convert Result to Decimal

10010110

1's Complement : 01101001

+1 : 01101010

01101010 = 106

Result = -106

Overflow Check

Carry into MSB (C7) = 1

Carry out of MSB (C8)= 0

$$\text{Overflow} = C7 \oplus C8$$

$$= 1 \oplus 0$$

$$= 1$$

$$\text{Stored Result} = -106$$

$$\text{Overflow Flag} = 1$$

Overflow Occurred

Question 2

Given

$$A = 1011$$

$$B = 1101$$

$$C_{in} = 0$$

Ripple Carry Adder Calculation

Bit 0

$$A_0 = 1 \quad B_0 = 1 \quad C_{in} = 0$$

$$\text{Sum}_0 = 0$$

$$\text{Carry} = 1$$

$$\text{Bit 1} \quad A_1 = 1 \quad B_1 = 0 \quad C_{in} = 1$$

$$\text{Sum}_1 = 0$$

$$\text{Carry} = 1$$

Bit 2

$$A_2 = 0 \quad B_2 = 1 \quad C_{in} = 1 \quad \text{Sum}_2 = 0$$

$$\text{Carry} = 1$$

Bit 3

$$A_3 = 1 \quad B_3 = 1 \quad C_{in} = 1 \quad \text{Sum}_3 = 1$$

$$\text{Carry} = 1$$

$$\text{Carry} = 1$$

$$\text{Sum} = 1000$$

$$\text{Final Result} = 11000$$

Carry Look-Ahead Adder Calculation

$$G_0 = A_0 \times B_0 = 1 \times 1 = 1$$

$$P0 = A0 \oplus B0 = 1 \oplus 1 = 0$$

$$G1 = A1 \times B1 = 1 \times 0 = 0$$

$$P1 = A1 \oplus B1 = 1 \oplus 0 = 1$$

$$G2 = A2 \times B2 = 0 \times 1 = 0$$

$$P2 = A2 \oplus B2 = 0 \oplus 1 = 1$$

$$G3 = A3 \times B3 = 1 \times 1 = 1$$

$$P3 = A3 \oplus B3 = 1 \oplus 1 = 0$$

Carry Calculation

$$C1 = G0 + P0C0$$

$$= 1 + (0 \times 0)$$

$$= 1$$

$$C2 = G1 + P1C1$$

$$= 0 + (1 \times 1)$$

$$= 1$$

$$C3 = G2 + P2C2$$

$$= 0 + (1 \times 1)$$

$$= 1$$

$$C4 = G3 + P3C3$$

$$= 1 + (0 \times 1)$$

$$= 1$$

Question 3

Given

Multiplicand (M) = -5

Multiplier (Q) = -3

Convert to 4-bit 2's Complement

$$+5 = 0101$$

$$-5 = 1011$$

$$+3 = 0011$$

$$-3 = 1101$$

Initialization

$$A = 0000$$

$$Q = 1101$$

$$Q-1 = 0$$

$$M = 1011$$

$$-M = 0101$$

Iteration 1

$$Q_0 \ Q-1 = 10$$

$$A = A - M$$

$$0000$$

$$+0101$$

$$0101$$

Arithmetic Right Shift

$$A \quad Q \quad Q-1$$

$$0101 \ 1101 \ 0$$

↓

$$0010 \ 1110 \ 1$$

Iteration 2

$$Q_0 \ Q-1 = 01$$

$$A = A + M$$

$$0010$$

$$+1011$$

$$1101$$

Arithmetic Right Shift

$$A \quad Q \quad Q-1$$

$$1101 \ 1110 \ 1$$

↓

1110 1111 0

Iteration 3

Q0 Q-1 = 10

A = A - M

1110

+0101

0011

Arithmetic Right Shift

A Q Q-1

0011 1111 0

↓

0001 1111 1

Iteration 4

Q0 Q-1 = 11

No Operation

Arithmetic Right Shift

A Q Q-1

0001 1111 1

↓

0000 1111 1

Final Product

AQ = 00001111

Question 4

Given

Dividend = 13

Divisor = 3

Dividend (Binary) = 1101

Divisor (Binary) = 0011

Initialization

A = 0000

Q = 1101

M = 0011

n = 4

Iteration 1

Left Shift (A,Q)

A = 0001

Q = 1010

A = A - M

0001

-0011

1110

A is Negative

Restore A

1110

+0011

0001

Q0 = 0

Iteration 2

Left Shift (A,Q)

A = 0011

Q = 0100

A = A - M

0011

-0011

0000

A is Positive

$Q_0 = 1$

Iteration 3

Left Shift (A,Q)

A = 0000

Q = 1010

A = A - M

0000

-0011

1101

A is Negative

Restore A

1101

+0011

0000

$Q_0 = 0$

Iteration 4

Left Shift (A,Q)

A = 0001

Q = 0100

A = A - M

0001

-0011

1110

A is Negative

Restore A

1110

+0011

0001

$Q0 = 0$

Final Output

Quotient (Q) = 0100 = 4

Remainder (A) = 0001 = 1

Question 5

Given

Number 1 = 1.5

Number 2 = 0.125

IEEE 754 Single Precision Format

Convert to Binary

$1.5_{10} = 1.1_2$

$0.125_{10} = 0.001_2$

Normalize the Numbers

1.5

$1.1_2 = 1.1 \times 2^0$

Sign = 0

Exponent = $127 + 0 = 127 = 01111111$

0.125

$0.001_2 = 1.0 \times 2^{-3}$

Sign = 0

Exponent = $127 - 3 = 124 = 01111100$

Exponent Alignment

Exponent Difference

$127 - 124 = 3$

Normalization

Result

$$1.1010000 \times 2^0$$

$$\text{Exponent} = 127$$

$$\text{Mantissa} = 1010000$$

IEEE 754 Result

$$\text{Sign} = 0$$

$$\text{Exponent} = 01111111$$

$$\text{Mantissa} = 10100000$$

Decimal Verification

$$1.5 + 0.125 = 1.625$$

Final Result

$$\text{Expected Result} = 1.625$$

$$\text{Obtained Result} = 1.625$$